

Further Maths

Core: Recursion & Financial Modelling

Investment Practice SAC

SESSION ONE

PROBLEM ONE

Kylie has invested \$1200 in a special savings account that pays simple interest, with the interest being calculated quarterly. Using V_n to represent the value of the investment after n quarters, the recurrence relation that models this investment situation is:

$$V_0 = \$1200 \quad V_{n+1} = V_n + 9$$

- (a) Write down, in full, a calculation process, based on the recurrence relation, that shows that the interest rate applying to this investment is 3.0 % per annum.

$$V_{n+1} = V_n + D \quad \text{Where the interest added in each quarter is } D = \frac{r}{100} \times V_0$$

$$\text{Solve } \left(9 = \frac{r}{100} \times 1200, r \right)$$

$$r = 3.0\% \text{ p.a.}$$

- (b) If the total value of Kylie's investment is \$1326.00, for how long has the investment been running?

$$\text{Using } V_n = V_0 + nD, \quad \text{where } V_n = \$1326.00, \quad V_0 = \$1200, \quad D = 9$$

$$1326 = 1200 + 9n$$

$$9n = 126$$

$$n = 14$$

This represents 14 quarters, or 3.5 years.

Alternatively, using the solve function in the CAS
Solve ($1326 = 1200 + 9n, n$)
 $n = 14$

PROBLEM TWO

Lynne has invested \$1500 in an account paying interest at 4.2 % per annum, compounded monthly. When asked, by her Mathematics teacher, to write a recurrence relation to model this investment, using V_n to represent the value of the investment after n months, she wrote the following:

$$(i) \quad V_0 = \$1500 \quad V_{n+1} = 0.042V_n$$

When her teacher told her that this recurrence relation was incorrect, she wrote the following:

$$(ii) \quad V_0 = \$1500 \quad V_{n+1} = 1.042V_n$$

When told that this recurrence relation was also incorrect, she eventually wrote the following, correct, recurrence relation:

$$(iii) \quad V_0 = \$1500 \quad V_{n+1} = 1.0035V_n$$

- (a) Explain why recurrence relation (i) is incorrect.

Compound interest is growing by a percentage from the existing 100%, that is adding the percentage of interest rate to 100% or $R = 1 + \frac{r}{100}$

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- (b) Explain why recurrence relation (ii) is incorrect.

The nominated interest rate is per annum, so it needs to be converted to monthly rate.

$$R = 1 + \frac{\frac{r}{12}}{100} = 1 + \frac{\frac{4.2}{12}}{100} = 1.0035$$

- (c) Use your calculator to determine recursively the value of the investment, correct to the nearest cent, each month, for the first six months, and complete the table below.

Month number	Calculation details	Value of investment
0		1500.00
1	1500×1.0035	1505.25
2	1505.25×1.0035	$(1510.5183...) = 1510.52$
3	$1510.5183... \times 1.0035$	$(1515.8051...) = 1515.81$
4	$1515.8051... \times 1.0035$	$(1521.1105...) = 1521.11$
5	$1521.1105... \times 1.0035$	$(1526.4343...) = 1526.43$
6	$1526.4343... \times 1.0035$	$(1531.7769...) = 1531.78$

- (d) Determine the amount of interest that Lynne has earned in the first six months, correct to the nearest cent.

$$\text{Interest} = \text{Amount earned} - \text{Principal} = \$1531.78 - \$1500.00 = \$31.78$$

- (e) If Lynne does not touch this investment for 18 months, how much will it be worth after this time, correct to the nearest cent?

Using the general formula would allow to calculate the direct value of V_{18}

$$V_n = R^n \times V_0, \text{ where } R = \left(1 + \frac{\frac{4.2}{12}}{100}\right) = 1.0035, n = 18, V_0 = 1500$$

$$V_n = 1.0035^{18} \times 1500 = \$1597.3645... \approx \$1597.36$$

- (f) If Lynne could earn 4.8 % per annum, compounded monthly, on her \$1500 investment, how much more interest, correct to the nearest cent, could she have earned over the eighteen months compared to earning interest at 4.2 % per annum, compounding monthly?

Using the general formula would allow to calculate the direct value of V_{18}

$$\text{Using } V_n = R^n \times V_0, \text{ where } R = \left(1 + \frac{\frac{4.8}{12}}{100}\right) = 1.004, n = 18, V_0 = 1500$$

$$V_n = 1.004^{18} \times 1500 = \$1611.7515... \approx \$1611.75$$

$$\text{Difference in interest} = \$1611.75 - \$1597.36 = \$14.39$$

SESSION TWO

An astute borrower is keen to minimise the cost of his home loan. He is about to borrow \$300 000 at 6.575 % p.a. interest, compounded monthly, and with monthly payments for twenty years. His accountant has advised him to pay every fortnight instead of monthly repayment; the term and total cost of his loan will be reduced.

- (a) Calculate the monthly payment required to amortise the loan, correct to the nearest cent.

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$20 \times 12 = 240$	6.575	300 000	?	0	12

Solve for Pmt = -2 249.9854... $\approx$ - \$2 249.99

- (b) Calculate the total paid over the term of the monthly payment loan.

Total paid = $20 \times 12 \times 2249.99 = \$539\,997.60$

Use Pmt = \$2249.99 to maintain the level of accuracy

- (c) Calculate the fortnightly payment.

Fortnightly payment = $\frac{12}{26} \times \$2249.99 = \$1\,038.4569 \approx \$1\,038.46$

Note, convert monthly payment to year then to fortnight

- (d) Calculate the number of fortnightly payments needed to repay the loan, rounded up to the next whole number.

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
?	6.575	300 000	- 1 038.46	0	26

Solve for N = 519.2306... $\approx$ 520 fortnightly payments.

Note at 519 fortnightly payments, the loan will not be paid off, therefore the number of payments need to be rounded up irrespective whether the first digit after the decimal point is less than 5 or not. In this case, the number is 2.

- (e) Calculate the total paid over the term of the fortnightly payment loan.

Total paid = $520 \times \$1\,038.46 = \$539\,999.20$

- (f) Calculate the number of years and months needed to pay the loan with fortnightly payments.

Number of years = $520 \div 26 = 20$ years

- (g) Calculate the total amount of money saved over the term of the loan by the borrower paying fortnightly.

Savings = Monthly - Fortnightly = $\$539\,997.60 - \$539\,999.20 = -\$1.60$

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- (h) Calculate the number of years and months of payments saved by paying fortnightly.

$$\text{Time saved} = 20 \text{ years} - 17 \text{ years } 2 \text{ months} = 2 \text{ years } 8 \text{ months}$$

- (i) the total amount paid each year by the monthly payment.

$$\text{Annual payment total} = 12 \times \$2250 = \$27\,000$$

- (j) the total amount paid each year by the fortnightly payment.

$$\text{Annual payment total} = 26 \times \$1125 = \$29\,250$$

- (k) Explain why this type of fortnightly loan is repaid quicker (and cheaper) than the monthly loan.,

The loan was able to be repaid more quickly by being repaid fortnightly – at 2 years and 8 months of payments being saved. The reason for the quicker repayment is due the increased frequency of the repayment. When repayments are made fortnightly, there are two extra payments made each year. This is because there are 26 fortnights compared to 24 half months in a year. REMEMBER!!! Increasing the repayment frequency will shorten the loan term.

SESSION THREE

PROBLEM ONE

- (a) Stephen has borrowed \$50 000 to invest in shares. He has taken out an interest-only loan with interest charged at 8.65 % per annum, compounded monthly. He will make monthly interest-only payments and pay back the full \$50 000 after three years. Calculate the amount of the monthly payment, correct to the nearest cent.

$$\text{Interest (payment)} = \frac{r}{100} \times V_0 = \frac{\frac{8.65}{12}}{100} \times 50000 = 360.41666... \approx \$360.42$$

- (b) If he makes \$65 000 from the sale of his shares after three years, calculate how much profit or loss he makes overall on the deal.

$$\text{Total expenses} = \text{Total payment} + \text{Principal} = 3 \times 12 \times \$360.42 + \$50\,000 = \$62\,975.12$$

$$\text{Profit} = \text{Income} - \text{expenses} = \$65\,000 - \$62\,975.12 = \$2024.88$$

- (c) Stephen now wants to borrow money to help finance a building project. If his bank can offer an interest-only loan at 8.5 % per annum interest, how much could he apply for (to the nearest five thousand dollars), if he can afford a **maximum** monthly payment of \$1500?

$$\text{payment} = \frac{r}{100} \times V_0$$

$$1500 = \frac{\frac{8.5}{12}}{100} \times V_0$$

$$V_0 = 211764.705 \dots \quad \text{Solve} \left(1500 = \frac{\frac{8.5}{12}}{100} \times V_0, V_0 \right)$$

He can apply for \$210 000 to keep the monthly repayment UNDER \$1500!!

- (d) What is the effective rate of interest, correct to two decimal places, that Stephen will be paying on his interest-only loan at 8.5 % per annum, compounded monthly?

$$r_{\text{effective}} = \left[\left(1 + \frac{r}{100} \right)^n - 1 \right] \times 100 = \left[\left(1 + \frac{\frac{8.5}{12}}{100} \right)^{12} - 1 \right] \times 100 = 8.83909 \dots \approx 8.84 \%$$

$$r_{\text{effective}} = \text{eff}(r, n) = \text{eff}(8.5, 12) = 8.83909 \dots \approx 8.84\%$$

PROBLEM TWO

- (a) Roger wants to establish a scholarship fund to support students at his old secondary college. He plans to invest \$125 000 in special bonds that pay 4.5 % per annum interest. If the interest is calculated and paid annually as the scholarship amount, how much money will be available each year to be awarded?

$$\text{Interest as scholarship yearly awards} = \frac{r}{100} \times V_0 = \frac{4.5}{100} \times 125000 = \$5625.00$$

- (b) Roger actually wants to have at least \$7500 available each year. How much money will he have to invest, to the nearest thousand dollars, at 4.5 % per annum to earn the required sum each year?

$$\text{Interest as scholarship awards} = \frac{r}{100} \times V_0$$

$$7500 = \frac{4.5}{100} \times V_0$$

$$\text{Solve} \left(7500 = \frac{4.5}{100} \times V_0, V_0 \right)$$

$$\text{Interest as scholarship awards} = \$166\,666.666 \dots$$

He will have to invest \$167 000 to meet the interest requirement.

(Note if he invests \$166 000, he will earn less than \$7500 per year – \$7470!)

PROBLEM THREE

Nick is self-employed as a sub-contract carpenter and has decided to set up a superannuation fund. He has started the fund with \$10 000 he has saved and invested it in an account earning 5.70 % per annum, compounded monthly. He has decided to add \$500 each month to the fund.

- (a) Use your calculator to determine recursively the value of the superannuation fund, correct to the nearest cent each month, for the first six months, and complete the table below.

Month number	Calculation details	Value of fund
0		10 000.00
1	$10\,000.00 \times 1.00475 + 500$	10 547.50
2	$10\,547.50 \times 1.00475 + 500$	(11 097.6006...) 11 097.60
3	$11\,097.6006... \times 1.00475 + 500$	(11 650.3142...) 11 650.31
4	$11\,350.3142... \times 1.00475 + 500$	(12 205.6532...) 12 205.65
5	$12\,205.6532... \times 1.00475 + 500$	(12 763.6300...) 12 673.63
6	$12\,763.6300... \times 1.00475 + 500$	(13 324.2573...) 13 324.26

- (b) If he can make monthly payments of \$500 for the next thirty years, how much will be in his fund when he retires?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$30 \times 12 = 360$	5.70	-10 000	-500	?	12

Solve for $FV = 529\,447.608...$

He will have \$529 447.61.

- (c) Nick believes that he can make a success of his business and he will be able to add more money to his superannuation fund over time. He plans to pay \$500 per month for the first ten years, \$600 per month for the next ten years and \$700 per month for the last ten years. In the first 20 years the superannuation fund offers Nick 5.70% per annum, however they are happy to increase it by 0.03% for the last 10 years for being a loyal customer. How much money will be in his superannuation fund after these thirty years?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$10 \times 12 = 120$	5.70	-10 000	-500	?	12

Solve for $FV = 98\,278.026...$

He will have \$98 278.03 after ten years.

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$10 \times 12 = 120$	5.70	-98 278.03	-600	?	12

Solve for FV = 270 290.488...

He will have \$270 290.49 after twenty years.

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$10 \times 12 = 120$	5.73	-270 290.49	-700	?	12

Solve for FV = 591 778.521...

He will have \$591 778.52 after thirty years.

PROBLEM FOUR

Jennifer now works from home as a freelance graphic designer after a number of years working in large organisations. She currently has \$125 000 in her superannuation account, but her accountant advises her that she will need to have at least \$650 000 in the account when she plans to retire in twenty years.

- (a) If her superannuation account is earning 5.4 % interest per annum, compounded monthly, how much will Jennifer have to add each month, rounded to the nearest dollar, to at least meet her intended goal in twenty years?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$20 \times 12 = 240$	5.4	-125 000	?	650 000	12

Solve for Pmt = -656.8208...

She will have to contribute \$657 per month which will give her \$650 000 in twenty years' time.

- (b) In twenty years' time when Jennifer goes into retirement, she wants to invest her savings into an annuity where it would provide her a regular fortnightly income of \$2 500 for the next twenty years. She needs to look for a financial institution what would be prepared to give her a good deal. At what interest rate does Jennifer need to negotiate with the financial institution in order for her to achieve this?

Show details of any Finance Solver calculations in the table provided below.

N	I(%)	PV	Pmt	FV	PpY, CpY
$20 \times 26 = 520$	?	-650 000	2 500	0	26

Solve for I = 7.9597...

He needs to secure and interest rate of 7.96% p.a.